Wavelet-Entropy Approach for Detection of Bridge Damages Using Direct and Indirect Bridge Records

Chengjun Tan[1]; Ahmed Elhattab, P.E., Aff.M.ASCE[2]; and Nasim Uddin, P.E., F.ASCE[3]

Abstract: Bridges as a key component of road networks require periodic monitoring to detect structural degradation for early warning. Early detection of loci and extent of structural flaws is essential to maintain safe bridge functioning. The elegant properties of continuous wavelet transform (CWT) in analyzing the signal in both time and frequency domains was the impetus to extensively employ this technique in structural health monitoring applications. However, the faint signature of structural damages in the recorded bridge responses curtails the merits of employing this technique. Furthermore, the selection process for the optimal CWT parameters that could capture signal discontinuities due to structural damages is an arbitrary process, which adds another level of uncertainty to wavelet transforms. This paper investigates compiling Shannon entropy to CWT to infer the loci and extents of structural damages in bridges. Entropy is a measure used to evaluate the randomness of the data. The more stochastic the data, the higher the entropy. In this article, Shannon entropy is used to associate a proper probability density function for the used wavelet to measure the entropy of the wavelet function at different scales. Implementing this technique facilitates selecting the optimal CWT parameters to better depict the signal; hence, identifying signal discontinuities becomes viable. The paper numerically investigates the fidelity of the proposed approach to identify bridge damages using midspan bridge response as well as using indirect records from a vehicle passing over the bridge. An implicit vehicle–bridge interaction (VBI) algorithm is used to mimic the vehicle–bridge interaction dynamics for different scenarios. **DOI: 10.1061/(ASCE)IS.1943-555X.0000577.** © *2020 American Society of Civil Engineers.*

Author keywords: Bridge health monitoring; Shannon entropy; Nondestructive evaluation; Wavelet transform; Wavelet entropy; Drive-by bridge inspection.

Introduction

The wavelet transform is highly sensitive to discontinuities in signals and for a long time has been used as a robust signal processing tool. Usually, it is used to identify sharp changes in continuous or discrete signals, which are often indicative of damage. Several studies have proven the robustness of wavelet-based algorithms in identifying the intensities and locations of structural damages, both numerically and using scaled laboratory tests (Hou et al. 2000; Liew and Wang 1998; Lu and Hsu 2002). Rucka and Wilde (2006) used Gaussian wavelet and a reverse biorthogonal wavelet to effectively detect the damage position of a one-dimensional beam and two-dimensional plate, respectively, without knowledge of either the structure characteristics or their mathematical model. In another field of study, Douka et al. (2003) and Khorram et al. (2012) established a correlation between the crack size and the highest magnitude for the coefficient of the wavelet. Hester and González (2012) proposed a wavelet-based approach using wavelet energy content, instead of the wavelet coefficient, which proved to be more sensitive to damages. In tandem with that, Poudel et al. (2007) and Shahsavari et al. (2017) used wavelet analysis in extracting beam mode shapes as a metric to localize damage.

Although the mode shapes (Poudel et al. 2007; Rucka and Wilde 2006; Shahsavari et al. 2017) or deflection (Douka et al. 2003; Nguyen and Tran 2010; Zhu and Law 2006) of a beam can be used as input signals for damage detection, on a bridge site, it may not be easy to measure them to the required level of accuracy and at high scanning frequencies (Hester and González 2012). Many kinds of research, therefore, have investigated the use of bridge's acceleration as the input signal (Cantero and Basu 2015; Hester and González 2012; Hou et al. 2000; Kaloop and Hu 2015; Tan et al. 2017b; Wang et al. 2016).

Recently, authors started to investigate invoking wavelet transform in the drive-by bridge inspection field. Drive-by bridge inspection is the science of using indirect readings from a transit inspection vehicle to monitor the structural deterioration of bridges. This field of research emerged when Yang et al. (Yang et al. 2004; Yang and Lin 2005) investigated the feasibility of extracting the fundamental bridge frequencies from the vehicle's response. The authors found that for a smooth road profile, the vehicle axle acceleration spectrum is dominated by the bridge frequency. Further, for different bridge damping ratios, they observed an apparent drop in the signal power as the damping ratio increases. The latter observation illuminates the fidelity of using indirect vehicle responses in monitoring bridge health condition. Following Yang's work, Nguyen and Tran (2010) applied a Symlet wavelet transform to the displacement response of a moving vehicle to infer the severity and the location of bridge cracks. The authors theoretically simulated the vehicle–bridge interaction dynamics using a

[1]Postdoc, Key Laboratory for Wind and Bridge Engineering of Hunan Province, College of Civil Engineering, Hunan Univ., Changsha 410008, China (corresponding author). ORCID: https://orcid.org/0000-0002-1106-7943. Email: tan1990@uab.edu

[2]Research Associate, ASCE/ASME Affiliate Member, Univ. of Alabama at Birmingham, 1075 13th St. S, Birmingham, AL 35205. ORCID: https://orcid.org/0000-0002-0154-5550. Email: aahattab@uab.edu

[3]Professor, Dept. of Civil, Construction, and Environmental Engineering, Chief Editor for ASCE Journal of Natural Hazards Review, Univ. of Alabama at Birmingham, 1075 13th St. S, Birmingham, AL 35205. Email: nuddin@uab.edu

Note. This manuscript was submitted on July 3, 2018; approved on July 6, 2020; published online on September 7, 2020. Discussion period open until February 7, 2021; separate discussions must be submitted for individual papers. This paper is part of the ***Journal of Infrastructure Systems***, © ASCE, ISSN 1076-0342.

four-degrees-of-freedom half-car and a cracked finite-element beam. The study revealed that the algorithm sensitivity extensively dropped when the vehicle speed increased. In similar work, Khorram et al. (2012) performed a numerical investigation to compare the performance of a direct and indirect wavelet-based damage detection approach. The deflection response was applied as input signals. They pointed out that the indirect method showed more efficacy on damage detection than the direct method. The indirect method can even detect small cracks with a depth of 10% of beam depth. Although their method showed good performance, the vehicle was idealized as a moving force, neglecting the vehicle inertial component.

Acceleration responses of moving sensors can be also applied to detect bridge damage (Cantero et al. 2019; Deng and Cai 2009; Fitzgerald et al. 2019; Tan et al. 2017a; Xu and Wu 2007). Fitzgerald et al. (2019) numerically investigated the feasibility of using bogie acceleration measurements from a passing train with wavelet analysis to detect the presence of bridge scour. McGetrick and Kim (2014) and McGetrick and Kim (2013) highlighted a damaged section used to induce a clear discontinuity in the acceleration response of the passing vehicle, which in return affects the wavelet coefficient. This feature allows damage to be identified and located only if the vehicle traverses the bridge with considerably low driving speed (2.5 m/s).

Successful use of wavelet transform requires proper selection for the wavelet scale. Wavelet scale plays an essential role because it outlines the required frequency band to observe or analyze the objective signal. Wavelet transform can be calculated at every possible scale. A high scale is associated with low frequency content, whereas a low scale corresponds to high frequency content (Hester and González 2012). The selection of the appropriate or optimum scale has always been a challenge for users in the application of wavelet transform (Chatterjee et al. 2006; Chen et al. 2009). In most structural health monitoring (SHM) applied with wavelet transforms, they aim to find and localize the discontinuity component of signals indicated by abnormal behavior of structures. It is well understood that not all wavelet coefficients give the desired results. Nguyen and Tran (2010) highlighted that only the particular scale associated with the wavelet coefficient can provide an identifiable result of damage. To date, there is no criterion for selecting scale values, and the process is currently done arbitrarily and subjectively.

This paper introduces a wavelet entropy theory (WET) to select the optimum scale that provides better detection of acceleration signal discontinuities. The article uses the signal entropy as a metric in the scale selection process. Entropy is one of the fundamental notions of science, which generally refers to disorder or uncertain (de Oliveira 2015). Shannon (2001) proposed the concept of information entropy (Shannon entropy), which is defined as a measure of the average information content produced by stochastic data as random events. The more stochastic the data, the higher the entropy. Therefore, Shannon entropy can be used to measure the energy distribution of stochastic data. For continuous wavelet transform (CWT), each wavelet shape can be associated with a probability density function, which allows defining the entropy of the wavelet function (de Oliveira and de Souza 2006). The objective of this paper is to compile Shannon entropy with wavelet transform to find the scale that has the lowest possible entropy for a particular signal. This scale will be the most representative scale for this signal because it has the lowest entropy. In tandem with that, the wavelet of this scale will be very sensitive to any signal disorder; therefore, damage detection becomes viable. This framework is introduced as wavelet entropy theory.

Wavelet Entropy Theory Background

Wavelet Transform

Wavelet analysis is a robust signal-processing technique, which relies on the introduction of an appropriate basis and a characterization of the signal by the distribution of amplitude in the basis. If the wavelet is required to form a proper orthogonal basis, it has the advantage that an arbitrary function can be uniquely decomposed and the decomposition can be inverted (Rosso et al. 2001). The wavelet is a smooth and rapidly vanishing oscillation function with outstanding localization in both the time and frequency domains. The continuous wavelet transform of a function $f(t)$ is defined as the integral transform of $f(t)$ with a family of wavelet functions $\Psi_{a,b}(t)$ for each a and b

$$\mathrm{WT}(a,b) = \int_{-\infty}^{+\infty} f(t)\Psi_{a,b}(t)dt \tag{1}$$

where

$$\Psi_{a,b}(t) = \frac{1}{\sqrt{a}}\Psi\left(\frac{(t-b)}{a}\right) \tag{2}$$

is called the daughter wavelet derived from the mother wavelet $\Psi(t)$ and satisfies the properties of $\int_{-\infty}^{+\infty}\Psi(t)dt = 0$ and $\int_{-\infty}^{+\infty}|\Psi(t)dt| < \infty$; and a and b = real-valued parameters, denoting the scale and translation parameters, respectively.

Shannon Entropy

In science, entropy is commonly associated with the amount of disorder or order; the higher the entropy, the greater the disorder (Blackburn 2005). The random event or unpredictability can be measured in terms of entropy. Claude Shannon, who is the father of entropy theory, gives a useful criterion for analyzing and comparing probability distribution (Shannon 2001). It allows measuring the amount of information for any distribution using the concept of Shannon entropy. Assuming a discrete random variable X with possible values $[x_1 \ldots x_n]$ and the probability distribution function $P\ (X)$, such that $\sum_{i=1}^{n} P(x_i) = 1$, the entropy can explicitly be defined as

$$E_{\mathrm{entropy}}(X) = -\sum_{i=1}^{n} P(x_i)\log_2 P(x_i) \tag{3}$$

where $P(x_i)\log_2 P(x_i) = 0$ when $P(x_i) = 0$.

The value $E_{\mathrm{entropy}}(X)$ is a function of the probabilities of the random variable, $P\ (X = x)$ in a sample (X) and not the random variable itself (x) (Bulusu and Plesniak 2015). This is in agreement with the fact that maximum entropy of a discrete random variable is achieved by a uniform distribution (de Oliveira and de Souza 2006; Shannon 2001). Now considering the example of a coin toss, not necessarily fair, assume the probability of heads is represented by $P(h)$, whereas the probability of tails is $1 - P(h)$. Following the previous equation, the entropy of this coin toss corresponding to $P(h)$ is plotted in Fig. 1. The entropy of the unknown result of the next toss of the coin is maximized when the coin is fair (practically, it is a uniform distribution). This is the situation of maximum uncertainty because it is most difficult to predict the outcome of the next toss. It is well understood that if the coin is not fair, there is less uncertainty because one side is more likely to come up than the other; the more unfair, the less uncertainty. Accordingly, the reduced uncertainty is quantified in a lower entropy. For instance, if $P(h)$ is 0.9, then the entropy of the coin toss becomes as low

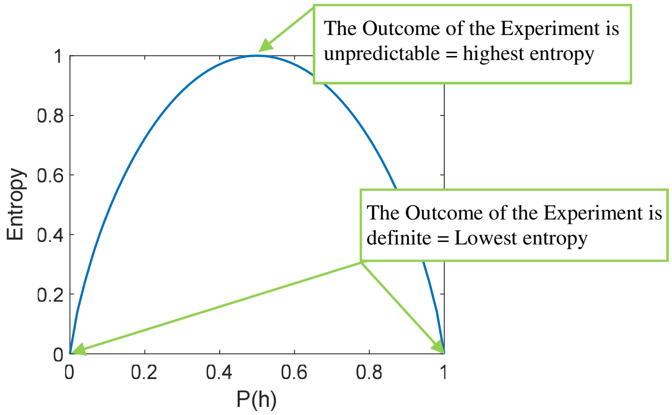

Fig. 1. The entropy of the coin toss experiment.

as 0.469. At an extreme, a coin toss using a coin that has two heads and no tails ($P(h) = 1$) has zero entropy because the coin will always come up heads, and the outcome can be predicted perfectly. As has been shown, the Shannon entropy is effective to measure the amount of uncertainty in the random data or event. It concluded once again that a system associated with higher entropy has more uncertainty with greater information (Saviotti 1988). In contrast, a system associated with lower entropy has an ordered state, consequently providing more stable, reliable, or useful information to some extent.

Algorithm of Optimum Wavelet-Scale Search

As mentioned previously, wavelet functions can be dilated into infinitely large or infinitesimally small scales (a). Therefore, a question that naturally arises by wavelet users is when or how to exit from the process of wavelet transformation; more specifically, what wavelet scale applies on the measured signal can effectively describe the interest of the user? For example, as shown in Fig. 2, the interest in the application of a wavelet is to detect the abrupt change in the contaminated sine function. Using the finer-scale daughter wavelet to approximate the objective signal highlights the noise component and leads to a number of redundant coefficients as noticeable distractions from the main results (Kailas and Narasimha 1999), whereas a greater-scale daughter wavelet will dilute the wave shape, resulting in inability to detect the partial changes of signals. How to select the optimal scale has always been the challenge in the application of wavelet transform. With the aid of the Shannon entropy measure, the optimal scale a (a_2) can be identified to depict signal discontinuity, which implies the presence of damage.

The energy of the wavelet coefficients at a particular scale a can be described as

$$E_{\text{energy}}(a) = \sum_{i=1}^{N} |\text{WT}(a,b)|^2 \tag{4}$$

where N = total number of wavelet coefficients.

The energy probability distribution of the wavelet coefficients is quantitatively described as

$$p_i = \frac{|\text{WT}(a,b)|^2}{E_{\text{energy}}(a)} \quad \text{with} \quad \sum_{i=1}^{N} p_i = 1 \tag{5}$$

Therefore, the wavelet Shannon entropy at each scale associated with the energy probability distribution can be calculated based on Eq. (3). Recall that the higher the entropy, the greater the disorder, so the optimal scale of daughter wavelet to approximate the measured signal should have the lowest Shannon entropy value, which can provide the most ordered situation to observe or most effectively represent the original signal. In the same vein, this scale will be the most sensitive to any signal disorders. In fact, a very ordered process could be thought of as a periodic monofrequency signal (signal with a narrow band spectrum). A wavelet representation of such a signal will be greatly resolved in one unique wavelet resolution level (particular scale a), and in consequence the Shannon entropy will be near zero or have a very low value (Ren and Sun 2008). However, in the previous example, the ideal ordered process does not exist. The optimal scale a_2 associated with lowest Shannon entropy provides the most ordered process in this case compared to others.

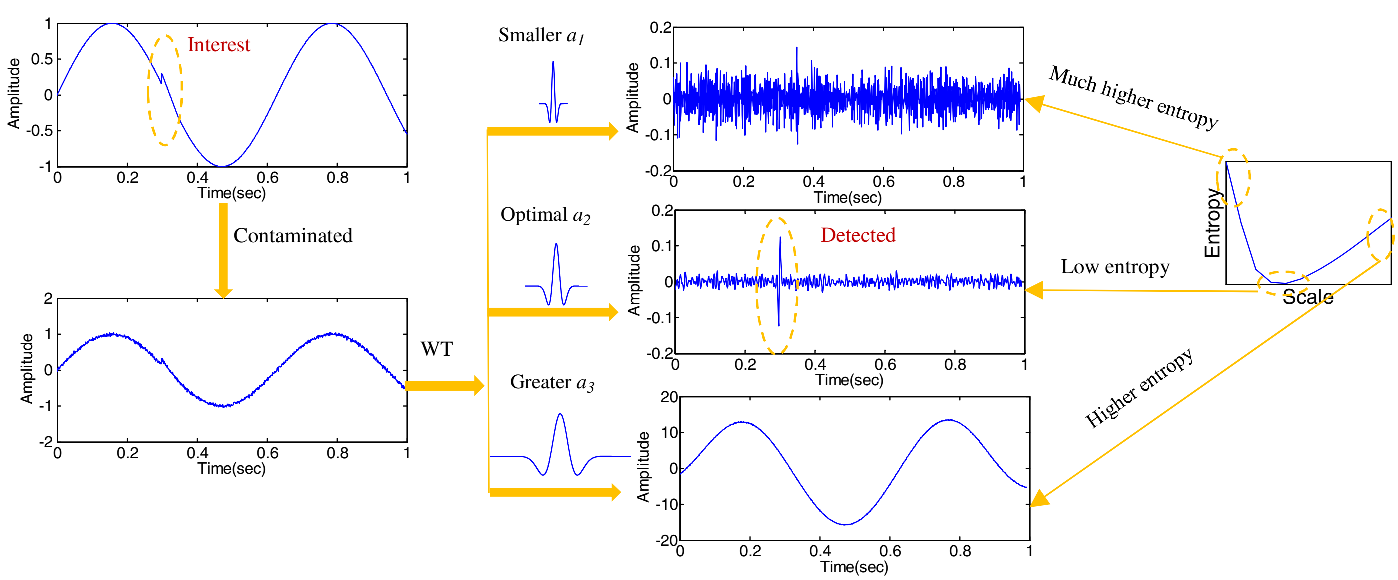

Fig. 2. Schematic of WET technique application.

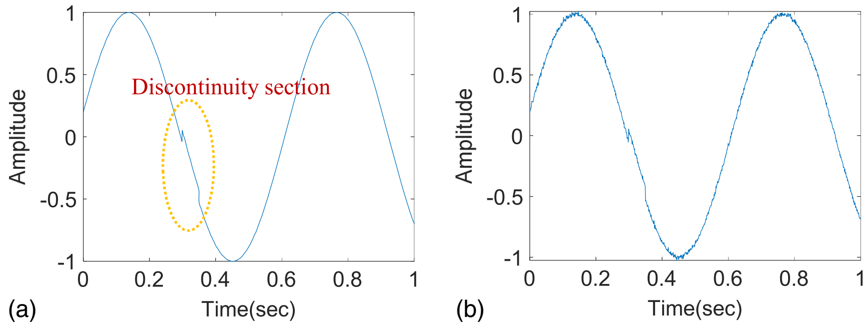

Fig. 3. Test signal: (a) original signal with discontinuity; and (b) contaminated signal.

From another perspective, a lower Shannon entropy value, which is attributed to lower signal uncertainty, translates to a higher concentration of the signal energy (Bulusu and Plesniak 2015). As shown in the example, the wavelet coefficients at both the smaller (a_1) and greater scales (a_3) have apparently dispersed energy distribution with higher Shannon entropy. Contrarily, the wavelet coefficients associated with the lowest Shannon entropy greatly concentrate the energy at the position at which the change happened. That is also the exact interest of detecting discontinuity or the damaged component from the measured signal in SHM applications.

The composition of wavelet transform and Shannon Entropy can be summarized in the following steps:

1. Acquisition of the measured data $X(t)$.
2. Applying WT on $X(t)$ at a series of scales from 1 to M with Eq. (1).
3. Calculating the wavelet energies at each scale and their energy probability distribution using Eqs. (4) and (5).
4. Calculating wavelet Shannon entropy at each scale to select the scale that is associated with lowest entropy value as the optimal scale for damage detection with Eq. (3).

Numerically Investigating the Fidelity of WET

In order to investigate the fidelity and effectiveness of the proposed approach, a simple example is given subsequently. Consider a sinusoidal signal $X(t) = \sin(10t + 0.2)$, where the sample size is 1,000 and the sampling frequency is 1,000 Hz. The discontinuity is present at the [300–350] sample window by adding a step function with 0.1 magnitude. The signal has been contaminated by white Gaussian noise of 40 dB noise intensity. Fig. 3 shows the original and corrupted signals.

Wavelet analysis has been applied with the mother wavelet of Gaussian 2 to identify the signal discontinuity section, and the contour map is illustrated in Fig. 4. In this plot, time and scale are represented by two mutually perpendicular horizontal axes, whereas wavelet coefficient is on the vertical axis. The variation of the signal power is depicted in the figure by the shade intensity. As shown, it is difficult to directly identify the discontinuity section from Fig. 4.

To examine the proposed approach, first the wavelet coefficient energy at each scale from 1 to 300 was calculated using Eq. (4). Then, the energy probability distribution of wavelet coefficients at each scale was obtained by applying Eq. (5). Finally, the wavelet entropy at each scale was computed with Eq. (3), and the results are shown in Fig. 5. The minimum entropy is at scale a of 4. The wavelet coefficient at scale 4 is taken out and illustrated in Fig. 6(b). It is evident in the figures that signal disorder has become clearly visible. The wavelet transform plot is rescaled, as shown in Fig. 6(a) for this scale, which shows two highlighted lines indicating the start and end points of signal discontinuity.

However, an apparent edge effect happened, which is a common challenge in the field of wavelet-based application. The edge effect may interfere with the decision of damage detection. To avoid the edge effect, the ideal, simple way is to acquire a much longer signal before and after the target component of the signal without introducing new discontinuity. However, that is not always possible. Alternatively, the target signal can also be extended with some techniques to decrease edge effects, such as zero padding, periodic extension, symmetric extension, data windowing, and so on (Torrence and Compo 1998). Fig. 7 plots the wavelet coefficient at scale 4 of the contaminated signal in Fig. 3(b) after using the zero padding technique. As shown, there is no clear edge effect that may compete with the effect of damage.

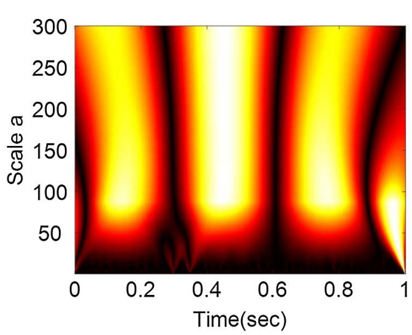

Fig. 4. Wavelet transform of the signal from scale 1 to 300.

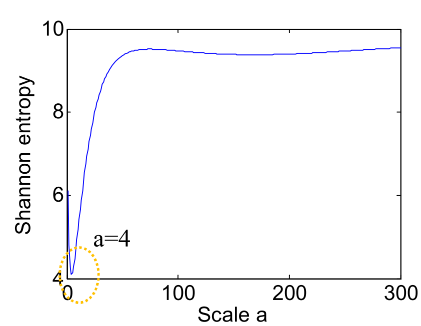

Fig. 5. Shannon entropy from scale 1 to 300.

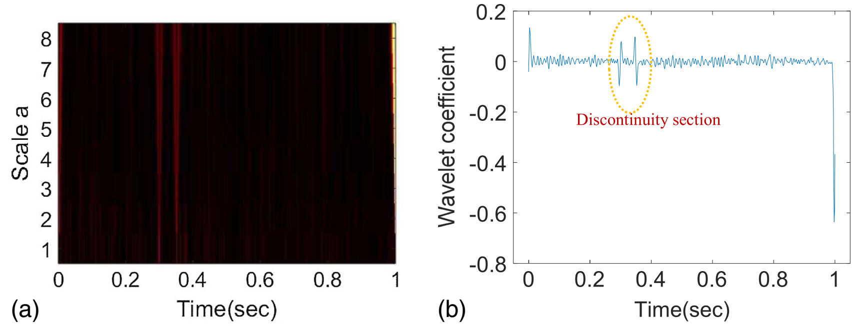

Fig. 6. CWT coefficients: (a) wavelet transform of signal from scale 1 to 8; and (b) wavelet coefficient at scale 4.

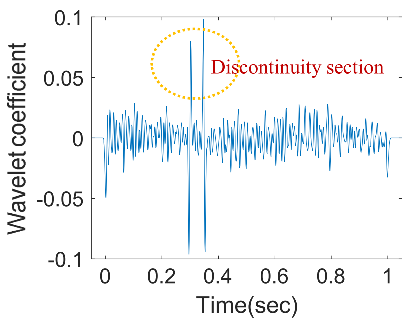

Fig. 7. Wavelet coefficient at scale 4 after zero padding.

Table 1. Vehicle and bridge properties

Vehicle properties		Bridge properties	
Property	Value	Property	Value
m_s	14,300 kg	Span	20 m
k_s	200 kN/m	Density	4800 kg/m^3
c_s	10 kN s/m	Width	4 m
m_a	700 kg	Depth	0.8 m
k_a	2,500 kN/m	Modulus	2.75×10^{10} N/m^2

Application of WET to Vehicle-Bridge Interaction Model

In this section, the feasibility of WET on localizing the bridge damage is investigated. An implicit vehicle-bridge interaction (VBI) solver algorithm was used to mimic the vehicle bridge interaction dynamics (Elhattab et al. 2016; Tan et al. 2019). Then, both of responses of the bridge (direct measurements) and passing vehicle (indirect measurements) were applied to the proposed algorithm to localize the bridge damage.

Application of WET to a Quarter-Car Model

First, the vehicle was modeled as a quarter-car model crossing a 20-m approach distance followed by a 20-m simply supported bridge (Fig. 8). The vehicle masses were represented using a sprung mass m_s and unsprung mass m_a, which represented the vehicle axle mass and body mass, respectively. The vehicle had axle mass bouncing degree of freedom u_a and sprung mass bouncing degree of freedom u_s. The properties of the quarter-car are listed in Table 1 (Cebon 1999; Harris et al. 2007). The dynamic interaction between the vehicle and bridge was implemented in MATLAB version R2018b. Unless otherwise mentioned, the scanning frequency was set as 1,000 Hz. The bridge fundamental frequencies were 2.07 and 8.38 Hz, whereas the vehicle frequencies were 0.57 and 10.33 Hz, respectively. Structural damage was modeled using the damage mode of Sinha et al. (2002). The damage level was defined as the ratio of the crack depth to the depth of the intact bridge. For instance, 0.1% or 10% damage level implies that the crack depth equals 0.08 m for a bridge with a depth of 0.8 m.

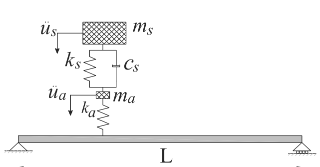

Fig. 8. VBI model.

The vehicle was simulated by crossing the approach and the bridge with a constant speed of 2.5 m/s. Fig. 9(a) illustrates the midspan acceleration response for the VBI model mentioned previously. The x-axis shows the normalized position of the vehicle axle on the bridge with respect to the bridge length (L) (0 and 1 when the axle is at the start and end of the bridge, respectively). The bridge is damaged at $0.35L$ (L = bridge span) where the damage level was set to 0.5. The road surface profile was not considered in this case. In the application of the wavelet, the choice of wavelet is important. Different wavelets with different characteristics may give different results. Hester and González (2012) pointed out that Gaussian 2 and Mexican Hat were the most successful at identifying the type of damage investigated in their research, which is the same type as this paper. Therefore, Shannon entropy was computed for wavelet scales with Gaussian 2 and was plotted as shown in Fig. 9(b). The optimal scale was found to be around 440, which will satisfy the minimal value of Shannon entropy. The wavelet coefficient at this particular scale was taken out and is shown in Fig. 9(c). The plot shows an evident peak around the crack position.

There are some traditional wavelet-based approaches for structural damage detection, which use a CWT contour map to represent the damage location directly (Hou et al. 2000; Kim and Melhem 2004). In these approaches, no method was proposed to select the range of scales that can identify the damage. However, it was found that the range of scale is important for damage detection. For example, the contour map associated with this case, as shown in Fig. 10(a), scale from 1 to 400, is not able to indicate any damage information, whereas the contour map as shown in Fig. 10(b), scale from 1 to 800, can show the difference happened at the bridge

J. Infrastruct. Syst., 2020, 26(4): 04020037

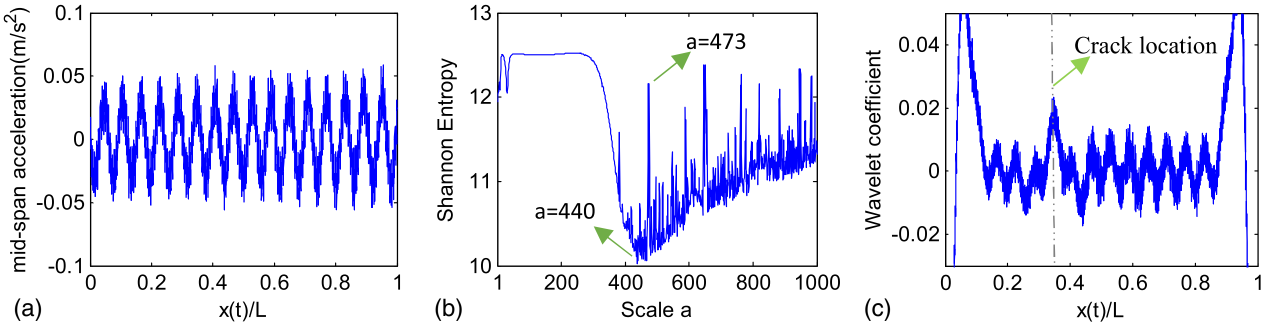

Fig. 9. WET on direct measurement: (a) bridge midspan acceleration; (b) Shannon entropy; and (c) wavelet coefficient.

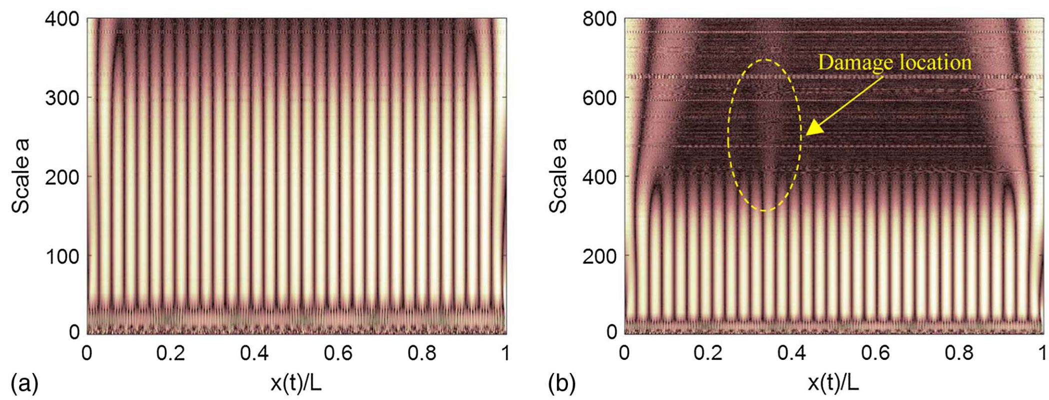

Fig. 10. Wavelet coefficients: (a) scales from 1 to 400; and (b) scales from 1 to 800.

damage location, even it is not obvious. It is clearly observed that Fig. 9(c) shows a better result than Fig. 10(b) for damage detection.

To further compare the proposed approach with the traditional approach, where CWT constants are randomly selected, in this case, the wavelet coefficient with Gaussian 2, at different manually picked scales, is presented in Fig. 11. It is evident from the plots that the scale that provides the best observation for the damage is where the entropy of the wavelet energy is minimal ($a = 440$). Lower scales ($a = 100 \rightarrow 300$) have the highest entropy [as depicted in Fig. 9(b)]; therefore, there is no sign of damage in their corresponding wavelet plots, as presented in Figs. 11(a–c). On the other hand, higher scales ($500 \rightarrow 700$) have lower entropy, and therefore damage is representable in their corresponding wavelet coefficient plots. The figure shows that the optimal scale that has the highest sensitivity to the signal disorder (or the signal discontinuity due to bridge damage) is where Shannon entropy is minimal. This finding establishes the fidelity of the proposed technique in selecting the appropriate wavelet scale to identify signal discontinuities present in the structure responses due to the existence of cracks or flaws. In addition, it is found that the wavelet coefficient with Mexican Hat can present a similar result indicating the crack location as well, where the optimal scale is 319.

Discrete wavelet transform (DWT) is another traditional approach used to detect structural damage existence and loci (Hou et al. 2000; Ovanesova and Suarez 2004). Generally, the details at Level 1 from the DWT are used for indicating structural damage, based on its obvious discontinuity. Hence, this acceleration in Fig. 9(a) was applied with five-level DWT to decompose, and all the details for each level are illustrated in Fig. 12. However, Level 1 detail, as well as the other details, showed no sign of any damage information. In this case, it showed that the bridge acceleration contained the damage information, but it was weak. It is difficult to extract the damage component information from the recorded bridge acceleration response. The proposed approach showed more effectiveness in detecting the damage than that based on DWT and a contour map.

Then, with the same case, the application of WET to indirect measurements was also investigated. The vertical acceleration of the quarter-car axle mass was used as the source signal. The signal is presented in Fig. 13(a). The acceleration response was analyzed with Gaussian 2. The wavelet scale that provides the minimum entropy was used to plot the wavelet coefficient. The location of the damage is evident in the plot, as shown in Fig. 13(c). Furthermore, the plot showed better identification (higher sensitivity) of structural damages when the vehicle response was used.

Application of WET with Random Traffic

Then, the scenario of the two quarter-cars was simulated in which another vehicle representing random traffic passed on the bridge. The traffic vehicle had the same characteristics as the test or instrumented vehicle, where the dynamic properties of m_s are 8,300 kg and the others are kept the same (the weight ratio of traffic vehicle to test vehicle is 0.6). In this case, the test vehicle and traffic vehicle had the same constant speed of 2.5 m/s, and the latter moved in front with 5 m between them.

Following the proposed procedures, the results of applying WET to direct or indirect measurements are shown in Figs. 14 and 15,

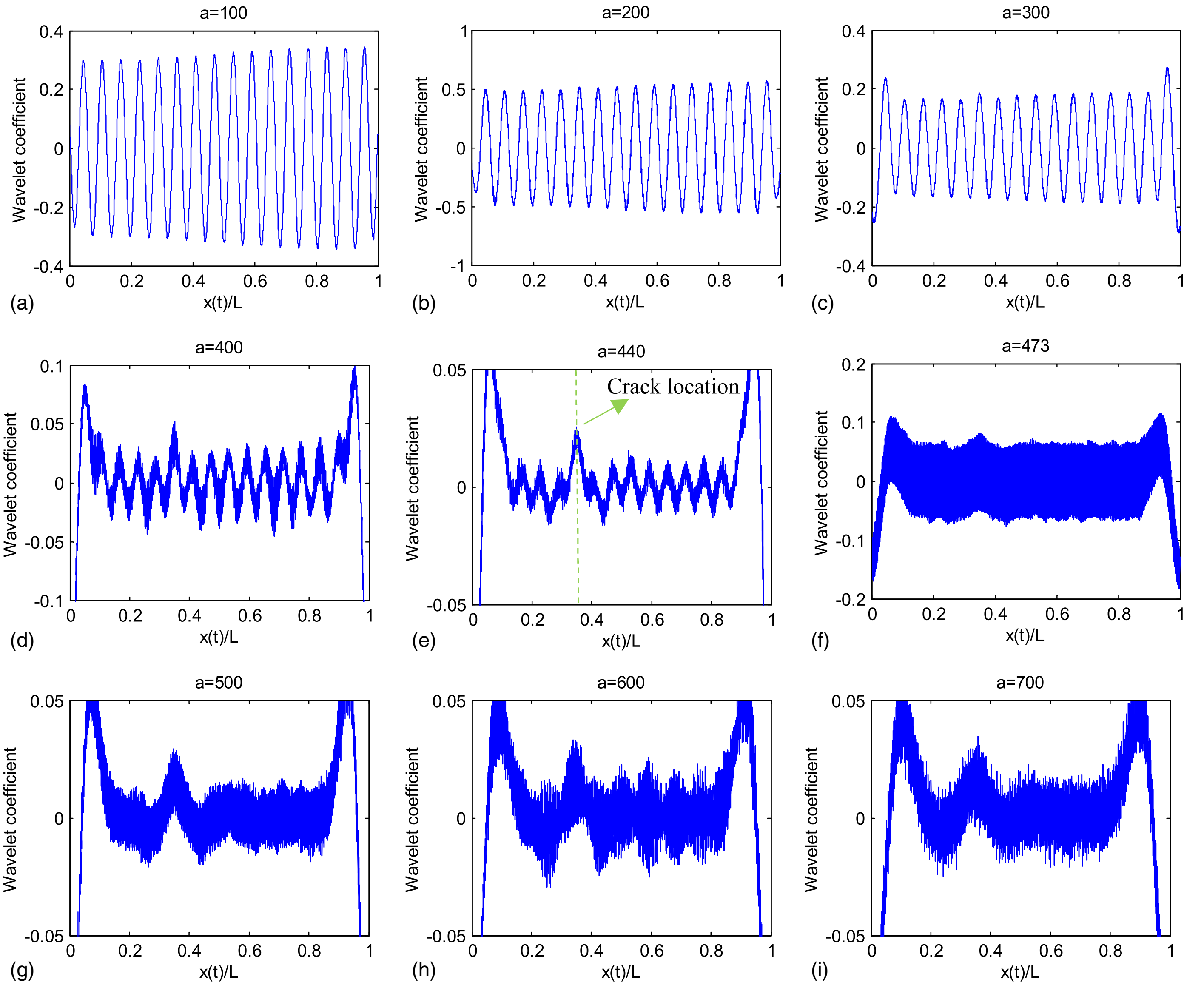

Fig. 11. Wavelet coefficients at different scales.

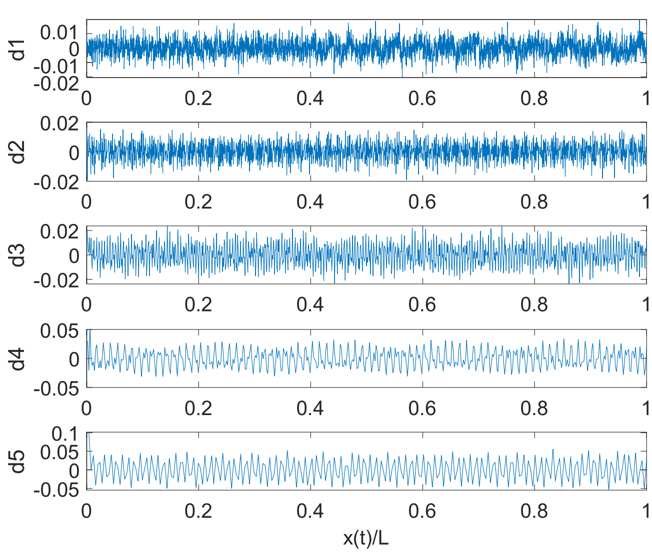

Fig. 12. Details in wavelet decomposition of acceleration response at five levels.

respectively. The x-axis shows the normalized position of the test vehicle on the bridge with respect to the bridge length. As shown, when the vehicle arrives or leaves the bridge, it introduced a discontinuity into the bridge and vehicle acceleration response that is captured by the wavelet coefficient. Both direct and indirect results can still show a visible peak at the bridge damage location on its optimum wavelet coefficient. In addition, the indirect result shows higher sensitivity to structural damages than that of direct measurement.

Application of WET to Half-Car Model

To further investigate the fidelity of the proposed WET on damage detection, a theoretical half-car model was adopted to represent the behavior of the vehicle, as shown in Fig. 16. This half-car model had four independent degrees of freedom, which correspond to vehicle body sprung mass bounce displacement u_s; sprung mass pitch rotation θ; and two tire displacements u_{aR} and u_{aF}, respectively. The vehicle body mass is represented by the sprung mass m_s and its moment of inertia is represented as I_s. The vehicle body connected to the tire via a combination of spring of linear stiffness

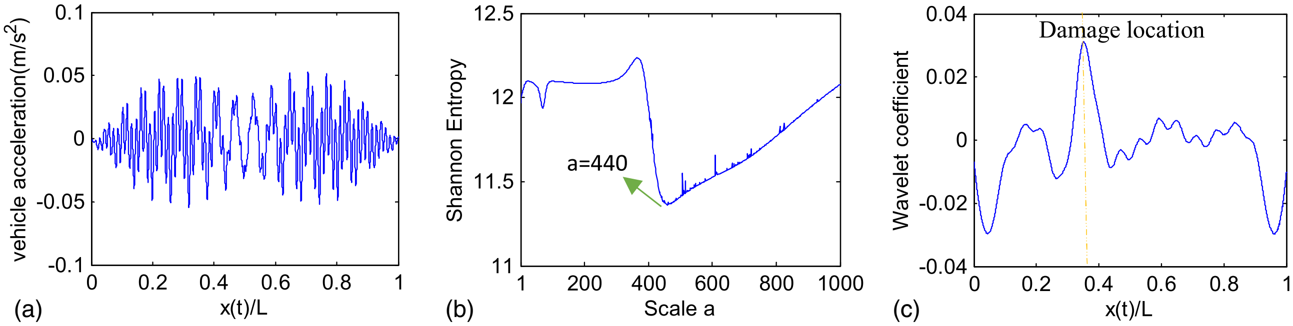

Fig. 13. WET on indirect measurement: (a) vehicle acceleration; (b) Shannon entropy; and (c) wavelet coefficient.

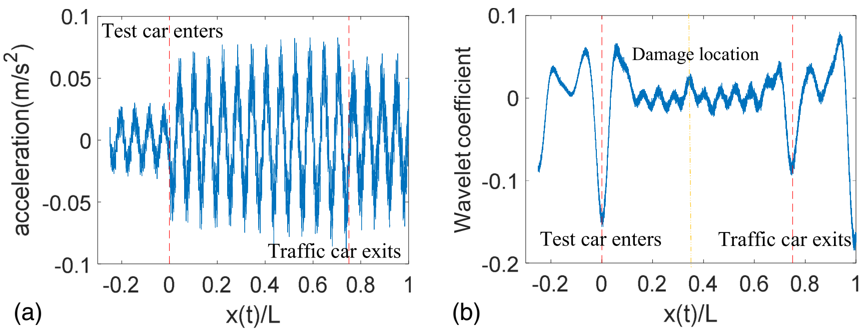

Fig. 14. WET on direct measurement with traffic: (a) Bridge midspan acceleration; and (b) wavelet coefficient.

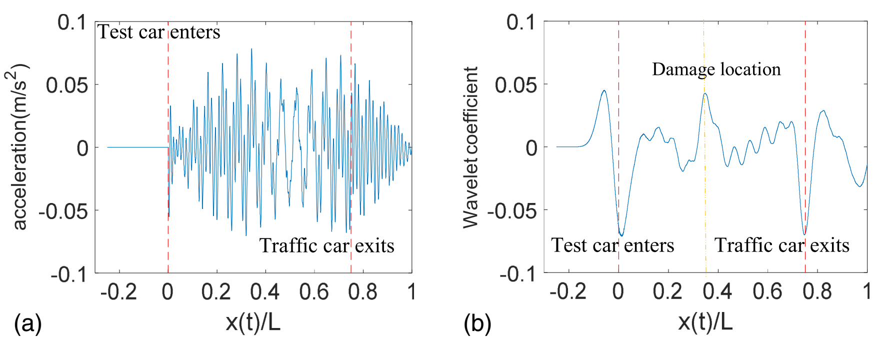

Fig. 15. WET on direct measurement with traffic: (a) vehicle acceleration; and (b) wavelet coefficient.

k_R or k_F and viscous dampers with damping coefficient C_R or C_F. The values of m_{aR} and m_{aF} represent the tire mass of each axle, and these masses contacted the road surface via springs of linear stiffness k_T. The values of $D1$ and $D2$ represent the distance of each axle to the vehicle's center of gravity. All of those instrumented vehicle properties are listed in Table 2.

The bridge properties were kept the same as the previous case, where the damage was still set at $0.35L$ at Level 0.5. Fig. 17(a) shows the bridge midspan acceleration when the half-car moved through the bridge with a constant speed of 2.5 m/s. The x-axis shows the normalized position of the vehicle rear axle on the bridge with respect to the bridge length. To detect the bridge damage, the proposed WET approach was applied to these signals. The optimum scale providing the minimum Shannon entropy value was achieved as 438, and the corresponding wavelet coefficient is illustrated in Fig. 17(b). Similar to multiple vehicles, when the vehicle axle arrived on or left the bridge, it also introduced a discontinuity into the bridge acceleration response. The wavelet coefficient still showed a small peak when the real axle passed over the bridge damage location. Principally, there should be two peaks caused by two axles when they pass over the damaged area. The main reason may be that the weight of the front axle is much less than the rear one. Therefore, one more case study was investigated, where the two axles had the same weight, namely $D1 = D2$ in the half-car model. Here the bridge damage location was set at $0.6L$ with same damage Level 0.5. Applying the same approach, the result is shown in Fig. 18, where two similar small peaks are presented. One of them was located at $0.6L$ in the normalized x-axis, whereas another one was located around $0.45L$, namely the locations when the corresponding axle passed over the damaged area.

The same case of the half-car model with the bridge damage at $0.35L$ was studied using indirect measurements. The proposed

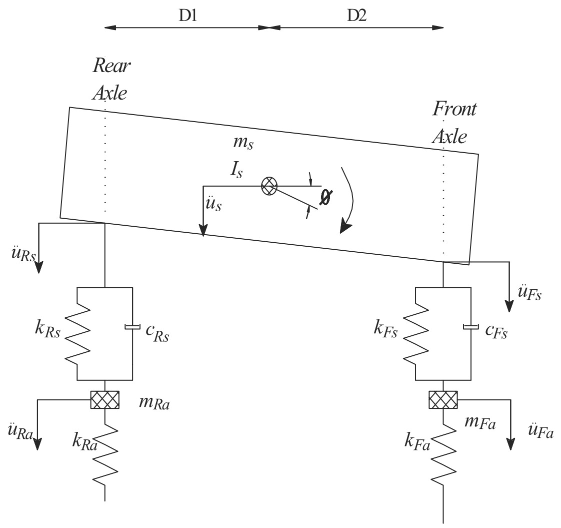

Fig. 16. Half-car model.

Table 2. Parameters of half vehicle model

Parameter	Value
m_s	16,600 kg
k_R	400 kN/m
c_R	8 kN s/m
m_{aR}	700 kg
k_T	1,750 kN/m
$D2$	1.05 m
I_s	95,765 kg m^2
k_F	400 kN/m
c_F	8 kN s/m
m_{aF}	700 kg
k_F	1,750 kN/m
$D1$	1.95 m

approach was applied to both axles' accelerations, and the results are illustrated in Fig. 19. The x-axis shows the normalized position of the corresponding axle on the bridge with respect to the bridge length. The optimum scales were 610 and 538 for the front and rear axle, respectively. Apparently, it is difficult to identify the bridge damage information from the front axle result, whereas the result of the rear axle points out a clear peak at the damage location. Therefore, this suggests using the rear axle (the more weighted one) response as the input signal to detect bridge damage associated with "drive-by" SHM. As in the quarter-car result, the wavelet coefficient of the axle presented better identification compared to direct measurement.

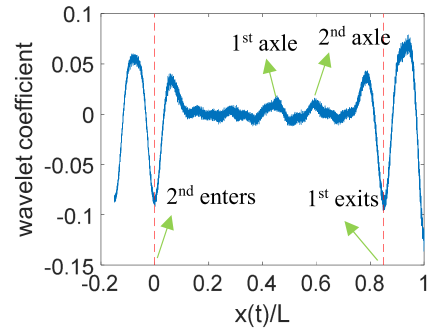

Fig. 18. Wavelet coefficient with half-car.

Parametric Investigation for the Fidelity of Using WET in Drive-by Bridge Inspection Concept

To investigate the fidelity of WET, the effects of the vehicle velocity, signal noise, damage loci, and severity were investigated. Only the indirect measurement was used in this section. A quantitative indicator was applied to evaluate the impact of each parameter on the approach sensitivity to damage detection. McGetrick and Kim (2014); Zhu and Law (2006) introduced a damage indicator as

$$\alpha = \frac{\log_2|\mathrm{WT}(s, u_d)|}{\log_2|\mathrm{WT}_o(s, u_d)|} \tag{6}$$

where u_d = damage location; s_0 represents a particular scale; and $WT(s, u_d)$ and $WT_o(s, u_d)$ = wavelet coefficients for the responses of the damaged and intact states at location u_d, respectively.

This damage indicator needs to be modified for this study for two main reasons. First, in each case, the particular scale might be different based on the entropy plot. Second, the peaks in wavelet coefficient plots do not usually present at the same position for each scale. Considering that there is only one major piece of damage to the structure, a damage indicator can be defined as

$$k = \frac{|\mathrm{WT}(s_0, u_{d1})|}{|\mathrm{WT}(s_0, u_{d2})|} \tag{7}$$

where u_{d1} = loci of the highest peak location; u_{d2} = loci of the second highest peak; s_0 represents the scale the provides minimum Shannon entropy; and $\mathrm{WT}(s_0, u_{d1})$ and $\mathrm{WT}(s_0, u_{d2})$ = their

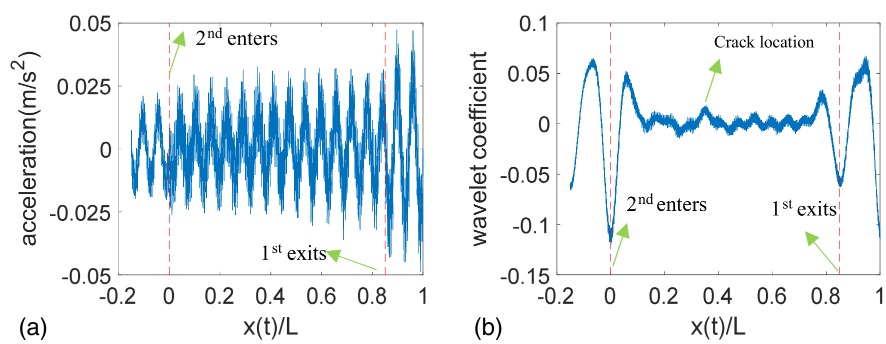

Fig. 17. WET on direct measurement with half-car: (a) bridge midspan acceleration; and (b) wavelet coefficient.

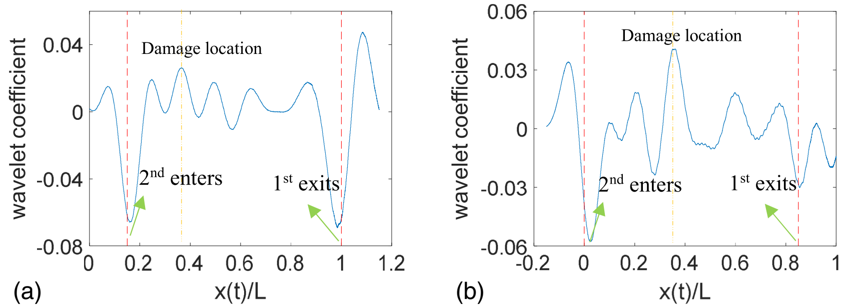

Fig. 19. Wavelet coefficient: (a) front axle; and (b) rear axle.

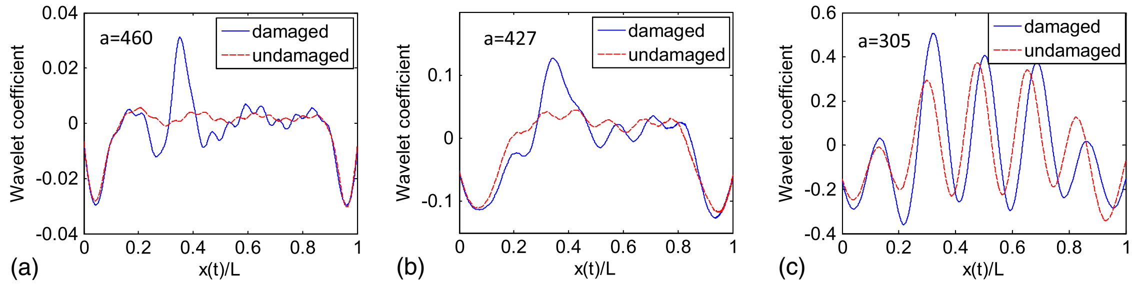

Fig. 20. Particular scale wavelet coefficients of different velocities: (a) 2.5 m/s; (b) 5 m/s; and (c) 7.5 m/s.

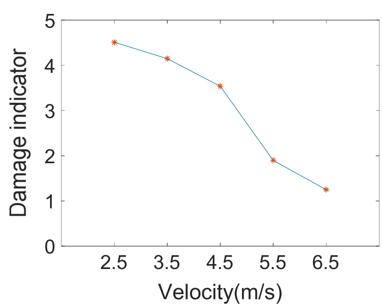

Fig. 21. Damage indicator at different vehicle velocities.

corresponding wavelet coefficients. Greater values for k reflect better identification of the damage.

Effect of Vehicle Speed

In this section, only the quarter-car was simulated. The damage was kept at $0.35L$, with a damage level of 0.5. Three speeds were investigated: 2.5, 5, and 7.5 m/s, referring to approximately 10, 20, and 30 km/h. Fig. 20 illustrates the optimal wavelet coefficient computed by the application of the proposed WET on axle acceleration records. The wavelet coefficient at the same scale for the undamaged response is also presented in the plots. Increasing the velocity decreased the predictability of the damage (k value). Fig. 21 shows the calculated damage indicator k for different vehicle velocities. With the increase of vehicle velocity, the decreased interaction time caused less information in the VBI model. That implies the "moving" sensor has lower resolution, allowing damage detection to be located less accurately. It was found that it was difficult to detect damage information when the vehicle velocity was greater than 10 m/s. This conclusion is consistent with McGetrick and Kim (2013, 2014).

Effect of Noise

The impact of the noise was investigated by adding white Gaussian noise to the vehicle responses. First, the signal of Fig. 13(a) was added to 60 dB white Gaussian noise, as shown in Fig. 22(a). By following the same procedure, the Shannon entropy and wavelet

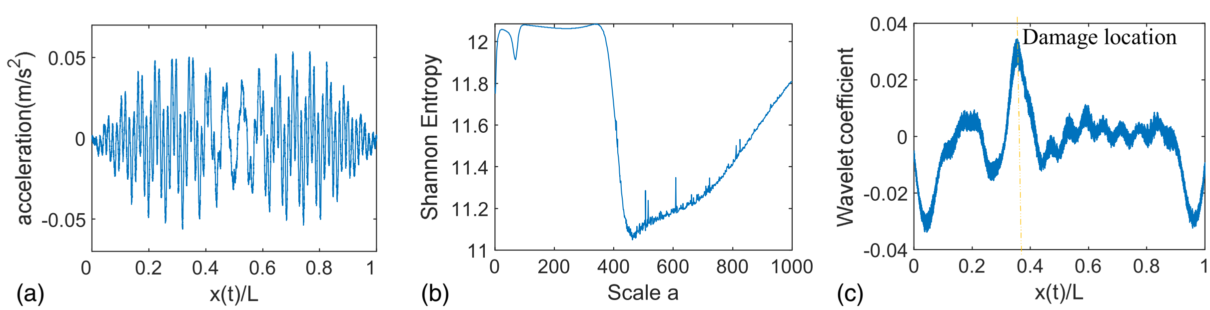

Fig. 22. WET on indirect measurement with 60-dB noise level: (a) polluted vehicle acceleration; (b) Shannon entropy; and (c) wavelet coefficient.

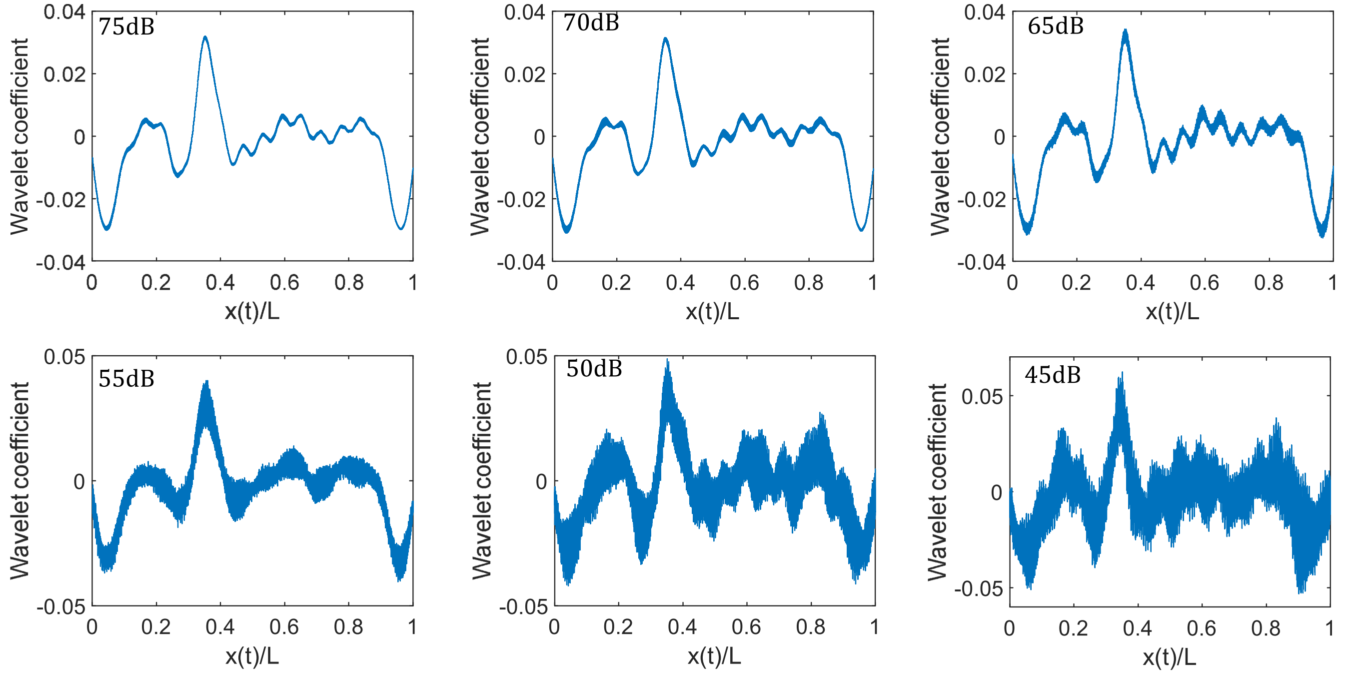

Fig. 23. Wavelet coefficients at different noise levels.

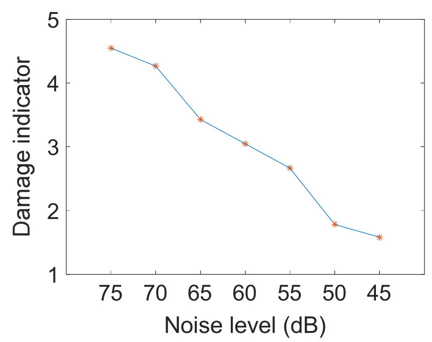

Fig. 24. Damage indicator at different noise levels.

coefficient used to identify the bridge damage were computed and are illustrated in Figs. 22(b and c), respectively. As shown, it was still able to localize the bridge damage where it happened at 0.35L.

To further investigate the effect of noise, the signal of Fig. 13(a) was reproduced by adding different noise intensities, and the results are presented in Fig. 23. The results reveal that the WET approach is insensitive to low noise contamination. The noise intensity was increased from 75 to 45 dB with a step of 5 dB. The damage indicator is plotted in Fig. 24. As noise intensity increased, the approach became less sensitive to structural damage. It was found that it was not able to identify bridge damage when the noise level reached 35 dB.

Different Damage Loci and Severity

Table 3 shows the values of the damage indicator for different damage locations and severities. The vehicle velocity was kept at 5.5 m/s for all cases. The damage index improved when the crack depth increased, as well as when the damage moved toward the bridge midspan. Fig. 25 shows wavelet coefficients for different

Table 3. Damage indicators identified at different loci and levels

	Level			
Damage loci	0.3	0.4	0.5	0.6
0.3L	1.87	2.26	2.50	2.94
0.35L	2.33	2.90	3.34	3.57
0.4L	3.19	4.33	5.04	5.64
0.45L	3.71	5.22	6.68	8.18

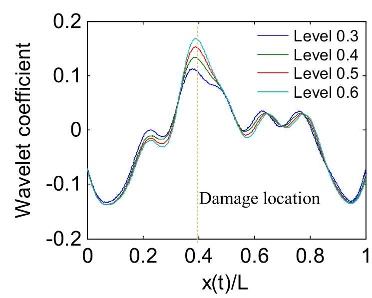

Fig. 25. Wavelet coefficients with different damage levels.

severity levels, where damage is located at 0.4L. The peak of the wavelet coefficient decreased with the decrease of damage severity.

Conclusions

Recently, wavelet analysis has frequently been used to detect structural damage by depicting signal discontinuities from structural dynamic responses. However, there is one major dilemma that halts the merits of this approach, namely the selection of the appropriate wavelet scale. This paper introduces the wavelet entropy theory, which provides a methodology for selecting the optimal wavelet scale by minimizing wavelet entropy. The approach is a step toward enhancing many existing wavelet-based damage detection techniques by introducing a methodology that can identify the wavelet scale that is most sensitive to structural damages. Entropy is referred to as the level of disorder or uncertainty of a set of random data. In the wavelet entropy theory, the entropy of the wavelet analysis at different scales is measured, where the scale that provides the least entropy is the most appropriate wavelet scale to represent the signal. Concurrently, the wavelet function at this scale will be the most sensitive function to any disorder in the signal, in other words, structural damages. The measurement fidelity of the WET approach was first verified using a simple sinusoidal signal. Afterward, the approach was expanded to a numerical vehicle bridge interaction problem including quarter-car and half-car models, where the measurements were taken from the bridge (direct measurements) and from the vehicle (indirect measurements). The results reveal a successful identification of structural damages for the two cases (the direct and indirect monitoring testbeds) when the signal-noise ratio (SNR) value is less than 35 dB. However, only a smooth profile was investigated in this research work. Future studies might include the effect of road roughness on the fidelity of the proposed WET method.

Acknowledgments

The National Science Foundation (No. NSF-CNS-1645863) sponsored this research. Any opinions, findings, and conclusions or recommendations expressed in this publication are those of the authors and do not necessarily reflect the view of the sponsors.

References

Blackburn, S. 2005. *The Oxford dictionary of philosophy.* Oxford, UK: Oxford University Press.

Bulusu, K. V., and M. W. Plesniak. 2015. "Shannon entropy-based wavelet transform method for autonomous coherent structure identification in fluid flow field data." *Entropy* 17 (12): 6617–6642. https://doi.org/10.3390/e17106617.

Cantero, D., and B. Basu. 2015. "Railway infrastructure damage detection using wavelet transformed acceleration response of traversing vehicle." *Struct. Control Health Monit.* 22 (1): 62–70. https://doi.org/10.1002/stc.1660.

Cantero, D., P. McGetrick, C. W. Kim, and E. Obrien. 2019. "Experimental monitoring of bridge frequency evolution during the passage of vehicles with different suspension properties." *Eng. Struct.* 187 (May): 209–219. https://doi.org/10.1016/j.engstruct.2019.02.065.

Cebon, D. 1999. *Handbook of vehicle-road interaction*. New York: Taylor & Francis.

Chatterjee, P., E. Obrien, Y. Li, and A. González. 2006. "Wavelet domain analysis for identification of vehicle axles from bridge measurements." *Comput. Struct.* 84 (28): 1792–1801. https://doi.org/10.1016/j.compstruc.2006.04.013.

Chen, S. L., J. J. Liu, and H. C. Lai. 2009. "Wavelet analysis for identification of damping ratios and natural frequencies." *J. Sound Vib.* 323 (1–2): 130–147. https://doi.org/10.1016/j.jsv.2009.01.029.

Deng, L., and C. Cai. 2009. "Bridge model updating using response surface method and genetic algorithm." *J. Bridge Eng.* 15 (5): 553–564. https://doi.org/10.1061/(ASCE)BE.1943-5592.0000092.

de Oliveira, H. M. 2015. "Shannon and Renyi entropy of wavelets." Preprint, submitted February 6, 2015. http://arxiv.org/abs/11502.01871.

de Oliveira, H. M., and D. de Souza. 2006. "Wavelet analysis as an information processing technique." In *Proc., Telecommunications Symp., 2006 Int.*, 7–12. New York: IEEE.

Douka, E., S. Loutridis, and A. Trochidis. 2003. "Crack identification in beams using wavelet analysis." *Int. J. Solids Struct.* 40 (13–14): 3557–3569. https://doi.org/10.1016/S0020-7683(03)00147-1.

Elhattab, A., N. Uddin, and E. Obrien. 2016. "Drive-by bridge damage monitoring using bridge displacement profile difference." *J. Civ. Struct. Health Monit.* 6 (5): 839–850. https://doi.org/10.1007/s13349-016-0203-6.

Fitzgerald, P. C., A. Malekjafarian, D. Cantero, E. J. OBrien, and L. J. Prendergast. 2019. "Drive-by scour monitoring of railway bridges using a wavelet-based approach." *Eng. Struct.* 191 (Jul): 1–11. https://doi.org/10.1016/j.engstruct.2019.04.046.

Harris, N. K., E. J. OBrien, and A. González. 2007. "Reduction of bridge dynamic amplification through adjustment of vehicle suspension damping." *J. Sound Vib.* 302 (3): 471–485. https://doi.org/10.1016/j.jsv.2006.11.020.

Hester, D., and A. González. 2012. "A wavelet-based damage detection algorithm based on bridge acceleration response to a vehicle." *Mech. Syst. Sig. Process.* 28 (Apr): 145–166. https://doi.org/10.1016/j.ymssp.2011.06.007.

Hou, Z., M. Noori, and R. S. Amand. 2000. "Wavelet-based approach for structural damage detection." *J. Eng. Mech.* 126 (7): 677–683. https://doi.org/10.1061/(ASCE)0733-9399(2000)126:7(677).

Kailas, S., and R. Narasimha. 1999. "The eduction of structures from flow imagery using wavelets Part I. The mixing layer." *Exp. Fluids* 27 (2): 167–174. https://doi.org/10.1007/s003480050341.

Kaloop, M. R., and J. W. Hu. 2015. "Stayed-cable bridge damage detection and localization based on accelerometer health monitoring measurements." *Shock Vib.* 2015: 1–11. https://doi.org/10.1155/2015/102680.

Khorram, A., F. Bakhtiari-Nejad, and M. Rezaeian. 2012. "Comparison studies between two wavelet based crack detection methods of a beam subjected to a moving load." *Int. J. Eng. Sci.* 51 (Feb): 204–215. https://doi.org/10.1016/j.ijengsci.2011.10.001.

Kim, H., and H. Melhem. 2004. "Damage detection of structures by wavelet analysis." *Eng. Struct.* 26 (3): 347–362. https://doi.org/10.1016/j.engstruct.2003.10.008.

Liew, K., and Q. Wang. 1998. "Application of wavelet theory for crack identification in structures." *J. Eng. Mech.* 124 (2): 152–157. https://doi.org/10.1061/(ASCE)0733-9399(1998)124:2(152).

Lu, C. J., and Y. T. Hsu. 2002. "Vibration analysis of an inhomogeneous string for damage detection by wavelet transform." *Int. J. Mech. Sci.* 44 (4): 745–754. https://doi.org/10.1016/S0020-7403(02)00006-1.

McGetrick, P., and C. Kim. 2014. "An indirect bridge inspection method incorporating a wavelet-based damage indicator and pattern recognition." In *Proc., 9th Int. Conf. on Structural Dynamics (EURODYN'14)*. Zurich, Switzerland: International Association for Bridge and Structural Engineering.

McGetrick, P. J., and C. W. Kim. 2013. "A parametric study of a drive by bridge inspection system based on the morlet wavelet." *Key Eng. Mater.* 569: 262–269. https://doi.org/10.4028/www.scientific.net/KEM.569-570.262.

Nguyen, K. V., and H. T. Tran. 2010. "Multi-cracks detection of a beam-like structure based on the on-vehicle vibration signal and wavelet analysis." *J. Sound Vib.* 329 (21): 4455–4465. https://doi.org/10.1016/j.jsv.2010.05.005.

Ovanesova, A., and L. E. Suarez. 2004. "Applications of wavelet transforms to damage detection in frame structures." *Eng. Struct.* 26 (1): 39–49. https://doi.org/10.1016/j.engstruct.2003.08.009.

Poudel, U. P., G. Fu, and J. Ye. 2007. "Wavelet transformation of mode shape difference function for structural damage location identification." *Earthquake Eng. Struct. Dyn.* 36 (8): 1089–1107. https://doi.org/10.1002/eqe.673.

Ren, W. X., and Z. S. Sun. 2008. "Structural damage identification by using wavelet entropy." *Eng. Struct.* 30 (10): 2840–2849. https://doi.org/10.1016/j.engstruct.2008.03.013.

Rosso, O. A., S. Blanco, J. Yordanova, V. Kolev, A. Figliola, M. Schürmann, and E. Başar. 2001. "Wavelet entropy: A new tool for analysis of short duration brain electrical signals." *J. Neurosci. Methods* 105 (1): 65–75. https://doi.org/10.1016/S0165-0270(00)00356-3.

Rucka, M., and K. Wilde. 2006. "Application of continuous wavelet transform in vibration based damage detection method for beams and plates."

J. Sound Vib. 297 (3–5): 536–550. https://doi.org/10.1016/j.jsv.2006.04.015.

Saviotti, P. P. 1988. "Information, variety and entropy in technoeconomic development." *Res. Policy* 17 (2): 89–103. https://doi.org/10.1016/0048-7333(88)90024-8.

Shahsavari, V., L. Chouinard, and J. Bastien. 2017. "Wavelet-based analysis of mode shapes for statistical detection and localization of damage in beams using likelihood ratio test." *Eng. Struct.* 132 (Feb): 494–507. https://doi.org/10.1016/j.engstruct.2016.11.056.

Shannon, C. E. 2001. "A mathematical theory of communication." *Mobile Comput. Commun. Rev.* 5 (1): 3–55. https://doi.org/10.1145/584091.584093.

Sinha, J. K., M. I. Friswell, and S. Edwards. 2002. "Simplified models for the location of cracks in beam structures using measured vibration data." *J. Sound Vib.* 251 (1): 13–38. https://doi.org/10.1006/jsvi.2001.3978.

Tan, C., A. Elhattab, and N. Uddin. 2017a. "Drive-by bridge frequency-based monitoring utilizing wavelet transform." *J. Civ. Struct. Health Monit.* 7 (5): 615–625. https://doi.org/10.1007/s13349-017-0246-3.

Tan, C., A. Elhattab, and N. Uddin. 2017b. "Wavelet based damage assessment and localization for bridge structures." In *Proc., 26th ASNT Research Symp.*, 228–240. Columbus, OH: American Society for Nondestructive Testing.

Tan, C., N. Uddin, E. J. OBrien, P. J. McGetrick, and C. W. Kim. 2019. "Extraction of bridge modal parameters using passing vehicle response." *J. Bridge Eng.* 24 (9): 04019087. https://doi.org/10.1061/(ASCE)BE.1943-5592.0001477.

Torrence, C., and G. P. Compo. 1998. "A practical guide to wavelet analysis." *Bull. Am. Meteorol. Soc.* 79 (1): 61–78. https://doi.org/10.1175/1520-0477(1998)079<0061:APGTWA>2.0.CO;2.

Wang, W., L. Deng, and X. Shao. 2016. "Number of stress cycles for fatigue design of simply-supported steel I-girder bridges considering the dynamic effect of vehicle loading." *Eng. Struct.* 110 (Mar): 70–78. https://doi.org/10.1016/j.engstruct.2015.11.054.

Xu, Z. D., and Z. Wu. 2007. "Energy damage detection strategy based on acceleration responses for long-span bridge structures." *Eng. Struct.* 29 (4): 609–617. https://doi.org/10.1016/j.engstruct.2006.06.004.

Yang, Y. B., and C. Lin. 2005. "Vehicle–bridge interaction dynamics and potential applications." *J. Sound Vib.* 284 (1): 205–226. https://doi.org/10.1016/j.jsv.2004.06.032.

Yang, Y. B., C. Lin, and J. Yau. 2004. "Extracting bridge frequencies from the dynamic response of a passing vehicle." *J. Sound Vib.* 272 (3): 471–493. https://doi.org/10.1016/S0022-460X(03)00378-X.

Zhu, X. Q., and S. S. Law. 2006. "Wavelet-based crack identification of bridge beam from operational deflection time history." *Int. J. Solids Struct.* 43 (7–8): 2299–2317. https://doi.org/10.1016/j.ijsolstr.2005.07.024.

J. Infrastruct. Syst., 2020, 26(4): 04020037